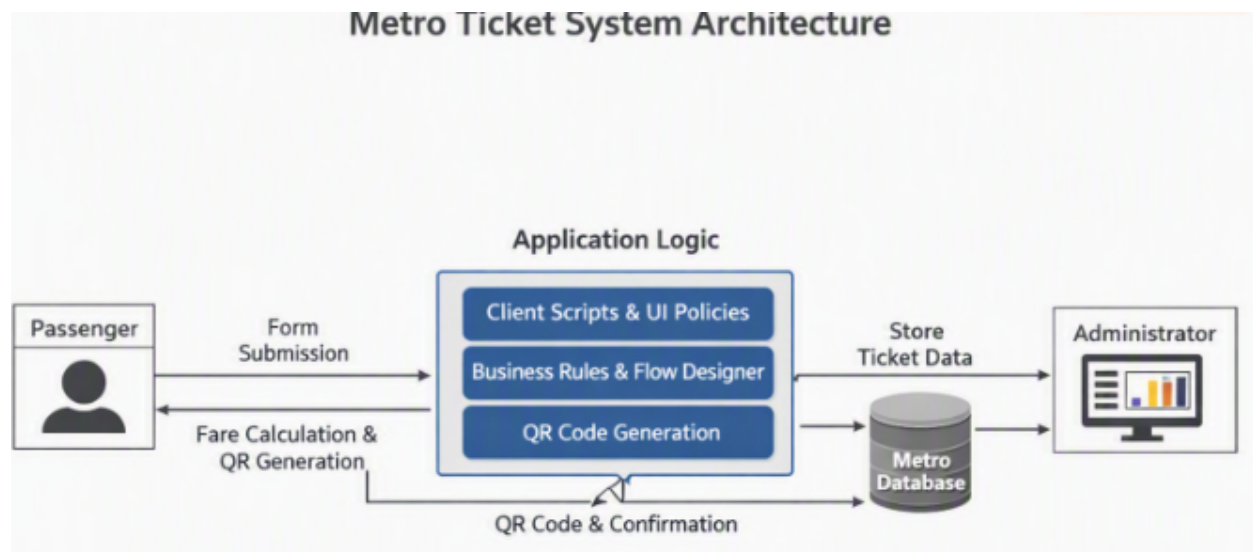


Project Design Phase-II Technology Stack (Architecture & Stack)

Date	
Student ID	23NU1A4221
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	4 Marks

Technology Stack (Architecture & Stack)

The Metro Ticket Generating System is implemented on the **ServiceNow cloud platform**. Passengers interact with the system through the **Service Portal and Service Catalog**. The booking form uses **Catalog Client Scripts and UI Policies** for dynamic behavior and validation. **Business Rules and Flow Designer** handle ticket processing and fare calculation, while the QR Server API generates the QR code.. Ticket information is stored in **ServiceNow tables**, and a **QR-code URL** is generated for the booked ticket. The QR code is displayed through a **Service Portal widget** and serves as digital proof for station verification.



- **Cloud-based Deployment:** Entire solution hosted on the ServiceNow cloud platform for scalability and reliability.
- **Dynamic Form Behavior:** Catalog Client Scripts and UI Policies ensure real-time validation and automated fare calculation.
- **Automated Ticket Processing:** Business Rules and Flow Designer handle ticket creation, fare computation, and QR code generation.
- **Secure Data Management:** Ticket and transaction details stored in ServiceNow tables with role-based ACLs.
- **Digital Ticketing Proof:** QR code generated via API and displayed in the Service Portal widget for station verification.

Table-1: Components & Technologies

S.No	Component	Description	Technology
1.	User Interface	Interface where passengers enter journey details and view QR ticket	ServiceNow Service Portal, Service Catalog, HTML, CSS, JavaScript
2.	Application Logic-1	Handles form behavior, field validation, conditional fields, dynamic inputs	ServiceNow Catalog Client Scripts, UI Policies, JavaScript
3.	Application Logic-2	Ticket processing and automated fare calculation	ServiceNow Business Rules, Flow Designer
4.	Application Logic-3	QR code generation and display	ServiceNow scripting, Service Portal Widget, QR Server API
5.	Database	Stores ticket, route/fare, booking, and payment details	ServiceNow Platform Database / Custom Tables
6.	Cloud Platform	Managed cloud storage for application records	ServiceNow Cloud Platform
7.	File Storage	No separate file storage required for core ticketing	Not Applicable
8.	External API-1	Generates QR code image from ticket/request identifier	QR Server API
9.	External API-2	No additional external API required	Not Applicable
10.	Machine Learning Mode	Not used in current implementation	Not Applicable
11.	Infrastructure	Hosts application, database, Service Portal, Catalog, automation services	ServiceNow Cloud Platform

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1.	Framework Usage	No external frameworks;ServiceNow provides platform + scripting environment	ServiceNow Platform, JavaScript
2.	Security Implementations	Role-based ACLs restrict access; authentication ensures secure transactions	ServiceNow ACLs, Roles, HTTPS
3	Scalable Architecture	Separation of UI logic, server logic, and data storage supports growth	Separation of UI logic, server logic, and data storage supports growth
4.	Availability	Hosted on ServiceNow cloud; accessible via Service Portal 24/7	ServiceNow Cloud Platform
5.	Performance	Client-side validation reduces unnecessary server requests, while fare calculation and ticket processing are handled efficiently	JavaScript, Catalog Client Scripts, Business Rules, Flow Designer